knowledge_transformer ◇◇◇◇◇◇

◇◇◇◇◇◇◇◇◇◇◇

1. ◇◇◇◇

1. ◇◇◇◇◇◇◇◇◇◇◇

◇◇ task_id API->>Q: ◇◇ conversion.handle_batch Q-->>W: ◇◇◇◇ W->>S: ◇◇◇◇◇◇◇◇ URL W->>P: ◇◇◇◇◇◇◇◇ P-->>W: ◇◇◇◇◇◇ W->>S: ◇◇◇ converted/task_id/ ◇◇ W->>API: ◇◇ results[] alt callback_url ◇◇◇ API->>CB: ◇◇ Webhook ◇◇ else ◇◇◇ API-->>API: ◇◇◇◇◇◇◇◇◇ end

    i. ◇◇◇◇◇◇◇◇◇ `task_id`◇
    ii. ◇◇◇◇◇◇ `conversion.handle_batch` ◇◇◇◇ Redis ◇◇◇
    iii. Worker ◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇ / ◇◇ URL◇◇
    iv. ◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇ `converted/{task_id}/` ◇◇◇
    v. ◇◇ `results[]` ◇◇◇◇◇◇◇ `callback_url` ◇◇◇◇◇

- **◇◇◇◇◇**

sequenceDiagram participant C as ◇◇◇ participant API as API ◇ participant R as Redis ◇◇/Result Backend participant W as Celery Worker participant M as ◇◇◇◇ participant Mon as Flower/Prometheus C->>API: POST /api/v1/convert API-->>C: 202 Accepted + task_id API->>R: ◇◇ conversion.handle_batch R->>W: ◇◇◇◇ W->>M: ◇◇◇◇◇/◇◇◇ W-->>R: ◇◇◇◇◇◇◇ alt callback_url ◇◇ W->>C: Webhook ◇◇ else ◇◇/◇◇ C->>R: ◇◇◇◇ C->>M: ◇◇◇◇ end Mon-->>API: ◇◇◇◇ Mon-->>W: ◇◇◇◇

    i. ◇◇◇◇◇ Pipeline Celery ◇◇◇◇ `/api/v1/convert` ◇◇◇◇◇◇◇ ◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇API ◇◇◇◇◇◇◇◇◇ Redis ◇◇◇
    ii. API ◇ HTTP 202 ◇◇ `task_id` ◇◇◇◇◇◇◇◇◇◇◇◇◇ ID ◇◇ Result Backend ◇◇◇◇◇◇
    iii. Celery Worker ◇◇◇◇◇◇◇◇◇◇◇◇◇ Webhook◇◇◇ Result Backend◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇ Webhook◇Result Backend ◇◇◇◇◇◇◇◇◇◇◇◇◇◇
    iv. Flower ◇ Prometheus ◇◇◇◇ API/Worker ◇◇◇◇◇◇◇◇◇◇◇◇ ◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇

4. ◇◇◇◇

- **◇◇◇◇**◇◇◇ Celery ◇◇◇◇◇◇◇◇◇◇◇◇
    - **◇◇**◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇ `priority` ◇◇◇ `prefetch_multiplier` ◇◇ worker ◇◇◇◇◇◇◇ `task_time_limit`◇ `soft_time_limit` ◇ `acks_late` ◇◇◇◇◇◇◇

- ◇◇◇◇◇API ◇◇◇ `files[]` ◇◇◇◇◇◇◇◇◇◇◇ SLA ◇◇ priority◇◇◇◇◇◇ `conversion.handle_batch`◇Worker ◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇ `max_tasks_per_child`◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇ `batch_size` ◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇

`--convert-to pdf:writer_pdf_Export`
- `tmpdir`
- `soffice`
- `--headless --norestore --nodefault`
- stderr
- `tmpdir`

- Inkscape CLI SVG/EPS/PDF PNG/JPEG/WebP
  - `--export-area-drawing`
  - `dpi`
  - Inkscape `--actions`
    -
      a. PDF/EPS Inkscape SVG
      b.
      c. `target_format` PNG/JPEG/WebP DPI/ `/tmp/rag_converter/<task>`
      d. `results[]`
      e. `converted/{task_id}/`

flowchart LR A[
SVG/EPS/PDF] --> B[
] B --> C[
PNG/JPEG/WebP] C --> D[
] D --> E[
converted/task_id/] E --> F[]

- FFmpeg `source_format` / `target_profile` Preset CRF ABR
    -
      a. `webm→mp4` MP4

x264 Baseline + CRF 23◇◇◇◇◇◇◇◇
AAC 128kbps◇

b. `wav→mp3` ◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇
16kHz/◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇

c. `mov→mp4` ◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇
H.265 ◇ H.264◇◇◇◇◇◇◇◇◇◇◇◇◇◇
◇◇

d. `gif→mp4/png` ◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇
◇ + scale ◇◇◇◇◇◇◇◇◇◇◇ MP4 ◇
PNG◇

5. ◇◇lib◇◇

- `rag_converter.plugins` ◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇
  - `ConversionInput/ConversionResult` ◇◇◇◇◇◇
    `source_format` ◇ `target_format` ◇ `input_path` / `input_url` / `object_key` ◇◇◇ `metadata` ◇◇◇◇◇◇
    `output_path` ◇ `object_key` ◇ `metadata` ◇
  - ◇◇◇◇◇◇◇

◇◇/◇◇	◇
`ConversionPlugin.convert(payload)`	`payload: ConversionInput`
`ConversionPlugin.describe()`	◇
`PluginRegistry.register(plugin_cls)`	`plugin_cls: Type[Convers`
`PluginRegistry.get(source, target)`	`source:str` , `target:str`
`PluginRegistry.list()`	◇
`load_plugins(module_names=None)`	`module_names: Iterable[s`
`read_plugin_module_file(path)`	`path: str

◇◇/◇◇	◇
`write_plugin_module_file(path, modules)`	`path: str

- `scripts/manage_plugins.sh` ◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇
- `rag_converter.monitoring` ◇◇◇ API/Worker ◇◇ Prometheus ◇◇◇

◇◇	◇◇◇◇
`ensure_metrics_server(port)`	`port:int` Prometheus ◇◇◇
`record_task_accepted(priority)`	`priority:str`
`record_task_completed(status)`	`status:str`
`collect_dependency_status(settings, celery_app)`	`settings:Settings` ◇ `celery_`
`_check_redis(settings)`	`settings:Settings`
`_check_minio(settings)`	`settings:Settings`
`_check_celery_workers(celery_app)`	`celery_app:Celery`

- `rag_converter.logging` ◇◇◇◇◇◇◇◇◇◇◇◇◇◇ JSON◇Trace/TaskID ◇ ◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇

◇◇	◇◇◇◇	◇◇/◇◇
`configure_logging(settings)`	`settings:LoggingSettings` ◇◇ level◇log_dir◇◇◇◇◇	◇◇◇◇◇◇◇◇◇◇ logging + structlog◇◇◇◇ Handler◇

- `pipeline_service` ◇◇◇◇◇ Celery ◇◇◇◇ Pipeline ◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇

◇◇/◇◇	
`CeleryConverterClient.submit_conversion_chain(files, priority)`	`files:List[D`
`submit_conversion_group(file_groups, priority)`	`file_groups:`
`submit_conversion_chord(file_batches, priority)`	`file_batches`
`get_result(task_id, timeout)`	`task_id:str`
`check_status(task_id)`	`task_id:str`
`quality_check_task(conversion_results)`	`conversion_r`
`post_process_task(checked_results)`	`checked_resu`
`aggregate_results_task(batch_results)`	`batch_result`

6. ◇◇◇◇◇◇◇◇◇◇/◇◇◇

◇◇/◇◇	◇◇
`POST /api/v1/convert`	Headers: `X-Appid`, `X-Key` ◇Body: `task_name`, `priority`, `cal` `storage{endpoint,access_key,secret_key,bucket}` ◇◇◇◇◇ `fi` `target_format`, `input_url` / `object_key` / `base64_data` ◇◇◇ `f`
`GET /api/v1/formats`	Headers: `X-Appid`, `X-Key`
`GET /api/v1/monitor/health`	Headers: `X-Appid`, `X-Key`
`GET /healthz`	◇
Celery ◇◇ `conversion.handle_batch`	`payload = {task_id, files[], priority, callback_url, stora`
Pipeline `submit_conversion_chain`	`files[]`, `priority`
Pipeline `submit_conversion_group`	`file_groups[][]`, `priority`
Pipeline `submit_conversion_chord`	`file_batches[][]`, `priority`

7. ◇◇◇◇

- ◇◇◇◇◇
  - ◇◇◇◇◇◇◇ `status=failed` + `reason` ◇◇
  - `task_time_limit=300s` ◇◇◇◇◇◇◇◇◇
  - ◇◇◇◇◇◇◇◇◇◇◇◇◇◇
  - ◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇
- ◇◇◇◇◇
  - ◇◇◇◇◇ + ◇◇◇◇◇◇◇◇
  - Webhook ◇◇◇◇◇◇◇◇◇
  - Result Backend/◇◇◇◇◇◇◇◇◇◇◇◇
  - Pipeline Celery ◇◇◇◇/◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇

8. ◇◇

- ◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇ Celery Chain◇◇
- ◇◇◇◇ `appid/key` ◇◇◇
- ◇◇ Redis◇◇◇◇◇◇LibreOffice◇Inkscape◇FFmpeg ◇◇◇◇◇◇
- ◇◇◇◇◇◇◇◇◇◇ `config/settings.yaml` ◇◇◇

9. ◇◇◇◇◇

◇◇◇	◇◇◇◇	◇◇/◇◇◇	◇◇◇◇
`doc`, `docx`, `ppt`, `xls`	`docx`, `pdf`, `html`	LibreOffice `soffice` ◇◇	◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇
`svg`, `eps`, `pdf`	`png`, `jpeg`, `webp`	Inkscape CLI	◇◇◇/◇◇◇◇◇◇◇◇◇◇◇◇◇◇
`gif`, `webp`	`png`, `mp4`	GIF/WebP ◇◇ + FFmpeg	◇◇◇◇◇◇◇◇◇◇◇◇◇◇
`wav`, `flac`, `ogg`, `aac`	`mp3`	FFmpeg Audio ◇◇	◇◇◇◇◇◇◇◇◇◇◇◇◇◇
`avi`, `mov`, `mkv`, `webm`, `mpeg`	`mp4`	FFmpeg Video ◇◇	◇◇◇◇◇◇◇◇◇◇◇◇◇◇
◇◇◇◇◇	◇◇◇◇◇	◇◇◇/◇◇◇◇	◇◇◇◇◇◇◇ CAD→PDF◇AI→SVG ◇◇

10. ◇◇◇◇

- ◇◇◇◇◇◇
  - Worker ◇◇◇◇ `/tmp/rag_converter/<task>` ◇◇◇◇◇◇◇◇◇
  - ◇◇◇◇ `converted/{task_id}/...` ◇◇◇◇◇◇◇ bucket ◇◇◇◇

11. ◇◇◇◇

- `config/settings.yaml` ◇◇◇◇◇◇◇◇◇◇◇ Redis/◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇

◇◇◇	◇◇◇◇	◇◇
`file_limits`	`default_max_size_mb`, `per_format_max_size_mb`, `max_files_per_task`	◇◇◇◇◇/◇◇◇◇◇◇◇◇◇◇◇◇
`logging`	`level`, `log_dir`, `max_log_file_size_mb`, `backup_count`	◇◇◇◇◇◇◇◇◇◇◇◇◇◇
`monitoring`	`prometheus_port`, `metrics_interval_sec`, `health_api` ◇	◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇
`minio`	`endpoint`, `access_key`, `secret_key`, `bucket`, `timeout`	◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇ `http://localhost:9000` / `minioadmin` / `minioadmin` / `qadata` ◇
`convert_formats`	`source`, `target`, `plugin`	◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇
`api_auth`	`required`, `app_secrets_path`,	API ◇◇◇◇◇◇◇◇◇◇◇

◇◇◇	◇◇◇◇	◇◇
	`header_appid`	
`celery`	`broker_url`, `result_backend`, `task_time_limit_sec`, `prefetch_multiplier`	◇◇◇◇◇◇◇◇◇◇
`rate_limit`	`enabled`, `interval_sec`, `max_requests`	API ◇◇◇◇◇
◇◇	`service_name`, `plugin_modules_file` ◇	◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇

- ◇◇◇◇◇◇◇ `RAG_` ◇◇◇◇◇ YAML◇◇
  ◇◇ `RAG_REDIS_URL` ◇ `RAG_MINIO_ENDPOINT` ◇ `RAG_API_AUTH_REQUIRED` ◇

◇◇◇◇	◇◇◇◇	◇◇
`RAG_REDIS_URL`	`celery.broker_url/result_backend`	◇◇ Celery Bro Result Backen
`RAG_MINIO_ENDPOINT`	`minio.endpoint`	◇◇◇◇◇◇◇◇ Docker ◇◇◇◇◇◇◇
`RAG_API_AUTH_REQUIRED`	`api_auth.required`	◇◇◇◇◇ API ◇◇◇
`RAG_FILE_LIMIT_MAX_SIZE`	`file_limits.default_max_size_mb`	◇◇◇◇◇◇◇◇
`RAG_PROM_PORT`	`monitoring.prometheus_port`	◇◇◇ Prometh ◇◇◇
◇◇ `RAG_*`	◇◇◇◇◇◇◇◇→◇◇◇◇	◇◇ `pydantic-setti` ◇◇◇◇◇◇ K Compose ◇◇◇

- CLI/□□□ `manage_plugins.sh` □ `make_key.sh` □□□□□□□□

□□	□□□□	□□/□
`manage_plugins.sh`	□□/□□/ □□□□□□□□□□□□ `config/plugins.yaml`	□□□□□□ `list/install/remove` □□□
`make_key.sh`	□□ API `appid/key` □□□ `secrets/appkeys.json`	□□ `--appid` □□ ID□□□□
`docker-start.sh` / `stop.sh`	□□□□/□□ Docker Compose □□	□□ `.env` □□□ `docker compose up/down` □□
`show_server.sh`	□□ API/Worker/ □□□□/Redis □□	□□□□□□□□□□□□□□□□

12. □□□□□□□

- `start_server.sh` □□□□□□□□ FastAPI□Celery Worker □ Flower□

□□	□□□□
FastAPI	`uvicorn rag_converter.app:app --host 0.0.0.0 --port ${API_PORT}`
Celery Worker	`celery -A rag_converter.celery_app.celery_app worker -l ${CELERY_LOG`
Flower	`celery -A rag_converter.celery_app.celery_app flower --port ${FLOWER`

- Celery Worker □ Pipeline Celery□□□ `.env` □□ `CELERY_BROKER_URL` □ `CELERY_RESULT_BACKEND` □ `MINIO_*` □□□□□□□□□□ □□□□□□□□□□□□□

◇◇	◇◇◇◇	◇◇
◇◇◇◇ Worker	`CELERY_BROKER_URL`, `CELERY_RESULT_BACKEND`, `CELERY_DEFAULT_QUEUE`, `CELERYD_PREFETCH_MULTIPLIER`, `TASK_TIME_LIMIT`, `MINIO_*`	◇◇ Broker/Backend◇◇◇◇◇/◇◇◇◇◇◇◇◇◇◇◇◇◇◇
Pipeline Celery	`BROKER_URL`, `RESULT_BACKEND`, `MINIO_ENDPOINT`, `MINIO_ACCESS_KEY`, `MINIO_SECRET_KEY`, `PIPELINE_QUEUE`	◇◇◇◇ Redis◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇

13. ◇◇◇◇

◇◇	◇◇◇◇/◇◇	◇◇
FastAPI	≥ 0.104	◇◇ REST API ◇◇◇◇
Celery	≥ 5.3	◇◇◇◇◇◇◇◇◇
Redis	≥ 7	◇◇ Celery Broker/Result Backend◇
◇◇◇◇	S3 ◇◇◇◇ MinIO◇	◇◇◇◇/◇◇◇◇◇
LibreOffice	◇◇ LTS	◇◇◇◇◇◇◇
Inkscape	≥ 1.3	◇◇◇◇◇◇◇◇
FFmpeg	≥ 5.0	◇◇◇◇◇/◇◇◇
Flower	≥ 1.2	Celery ◇◇ UI◇
Prometheus	≥ 2.46	◇◇◇◇◇◇◇◇

- Python ◇◇◇ `pyproject.toml`◇Pipeline ◇◇◇ `requirements.txt` ◇◇ README ◇◇◇◇

14. ◇◇◇◇◇◇

- ◇◇◇◇◇pytest + HTTPX◇Celery test worker◇◇◇◇◇ mock ◇◇◇◇◇◇
- ◇◇◇◇◇◇◇◇◇◇◇◇◇/◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇Webhook ◇◇◇◇◇◇◇ ◇◇◇◇ Pipeline Chain/Group/Chord ◇◇◇
- ◇◇◇◇◇◇◇◇◇

◇◇◇	◇◇◇ → ◇◇◇◇	◇◇ (MB)	◇◇
`doc_sample_small.doc`	`doc → docx`	0.05	◇◇◇◇◇◇◇◇◇◇◇◇◇ Chain◇◇
`doc_sample_pdf.doc`	`doc → pdf`	0.05	◇◇◇◇ PDF◇◇◇ sof
`html_inline_base64.json`	`html(base64) → pdf`	0.01	◇◇◇◇◇◇base64_d ◇◇◇◇◇◇
`ppt_marketing.ppt`	`ppt → pdf`	48	◇◇◇◇◇◇◇◇◇◇◇◇◇
`svg_logo.svg`	`svg → png`	0.8	◇◇/◇◇◇◇ + Pipelin
`gif_banner.gif`	`gif → mp4`	25	◇◇◇◇◇◇◇◇◇◇◇◇◇
`webp_large.webp`	`webp → png`	15	◇◇◇◇◇◇◇◇◇◇◇◇◇
`wav_podcast.wav`	`wav → mp3`	180	◇◇◇◇◇◇◇◇◇◇◇◇◇
`flac_archive.flac`	`flac → mp3`	220	◇◇◇◇◇◇◇ `FILE_T`
`mov_trailer.mov`	`mov → mp4`	480	◇◇◇◇◇◇◇◇◇◇ 500 Pipeline Chord◇
`mkv_fail.mkv`	`mkv → mp4`	300	◇◇◇◇◇◇◇◇◇◇◇◇◇
`invalid_format.bin`	`bin → docx`	1	◇◇◇◇/◇◇◇◇◇◇◇ ◇◇◇◇
`auth_test.docx`	`docx → pdf`	2	API key ◇◇◇◇◇◇◇
`webhook_payload.json`	-	-	Webhook ◇◇◇◇◇ 5

- ◇◇◇◇◇**Mock** ◇◇◇◇

◇◇	◇◇	◇◇	**Mock ◇◇**
UT-01	API ◇	API ◇◇◇◇◇	Mock Redis + Celery◇HTTPX client
UT-02	API ◇	◇◇◇◇◇◇	Mock `file_limits` ◇◇
UT-03	Worker	◇◇◇◇◇◇◇◇	Mock MinIO SDK ◇ `S3Error`
UT-04	Worker	◇◇◇◇◇◇	Mock ◇◇ `convert` ◇◇ `/tmp/out.docx`
UT-05	Worker	◇◇◇◇	Mock Celery ◇◇
UT-06	◇◇	API Key ◇◇	Mock `make_key.sh` + `secrets/appkeys.json`
UT-07	◇◇	Webhook ◇◇/◇◇	Mock `requests.post`
UT-08	Pipeline	Chain ◇◇◇	Mock `conversion/quality_check/post_process`
UT-09	Pipeline	Group + Result Backend	Mock `AsyncResult`
UT-10	◇◇	◇◇◇◇	Mock Redis/MinIO/Celery Ping
UT-11	◇◇◇◇	Logging ◇◇	Mock `structlog.configure`
UT-12	Worker	base64 ◇◇◇◇	Mock/◇◇ MinIO
UT-13	Worker	◇◇◇◇◇◇	Mock MinIO ◇◇◇

- **◇◇/◇◇/◇◇◇◇◇**
  - **◇◇◇◇**◇◇ UT-01/UT-07 ◇◇◇◇◇◇◇◇◇◇◇pytest-xdist + Celery test worker◇◇◇◇ `queue_depth` ◇ `prefetch_multiplier`

◇◇◇◇◇◇◇
    - ◇◇◇◇◇◇ `file_limits`◇◇◇ 1KB◇◇◇ 500MB◇◇◇◇◇◇1 ◇◇◇ 10◇◇◇◇◇◇◇low/high◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇
    - ◇◇◇◇◇◇◇ UT-03/UT-05/UT-06◇◇◇◇◇◇◇◇◇◇◇◇◇ 5xx◇ Webhook ◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇
- ◇◇◇◇◇
    - ◇◇ HTTP/◇◇◇◇
    - ◇◇ Redis ◇◇◇◇◇◇◇◇◇◇◇◇◇◇
    - ◇◇ Prometheus ◇◇◇◇◇◇◇◇◇/◇◇◇◇◇
- **◇◇◇◇**◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇/◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇◇

◇◇◇	**HTTP ◇◇**	◇◇◇	◇◇	◇◇◇◇
`ERR_AUTH_INVALID`	401	4011	◇◇◇◇◇appid ◇ key ◇◇	◇◇◇◇ key ◇◇ key◇
`ERR_FILE_TOO_LARGE`	400	4201	◇◇◇◇◇◇	◇◇ `per_format_max` `flac_archive.fl`
`ERR_BATCH_LIMIT_EXCEEDED`	400	4202	◇◇◇◇/ ◇◇◇◇	◇◇ `max_files_` `max_total_uploa`
`ERR_FORMAT_UNSUPPORTED`	400	4203	◇◇◇◇◇◇	◇◇◇◇◇◇◇ `invalid_format`
`ERR_TASK_FAILED`	500	5001	◇◇◇◇◇◇	◇◇◇◇◇◇◇/ ◇◇◇◇◇◇◇◇◇